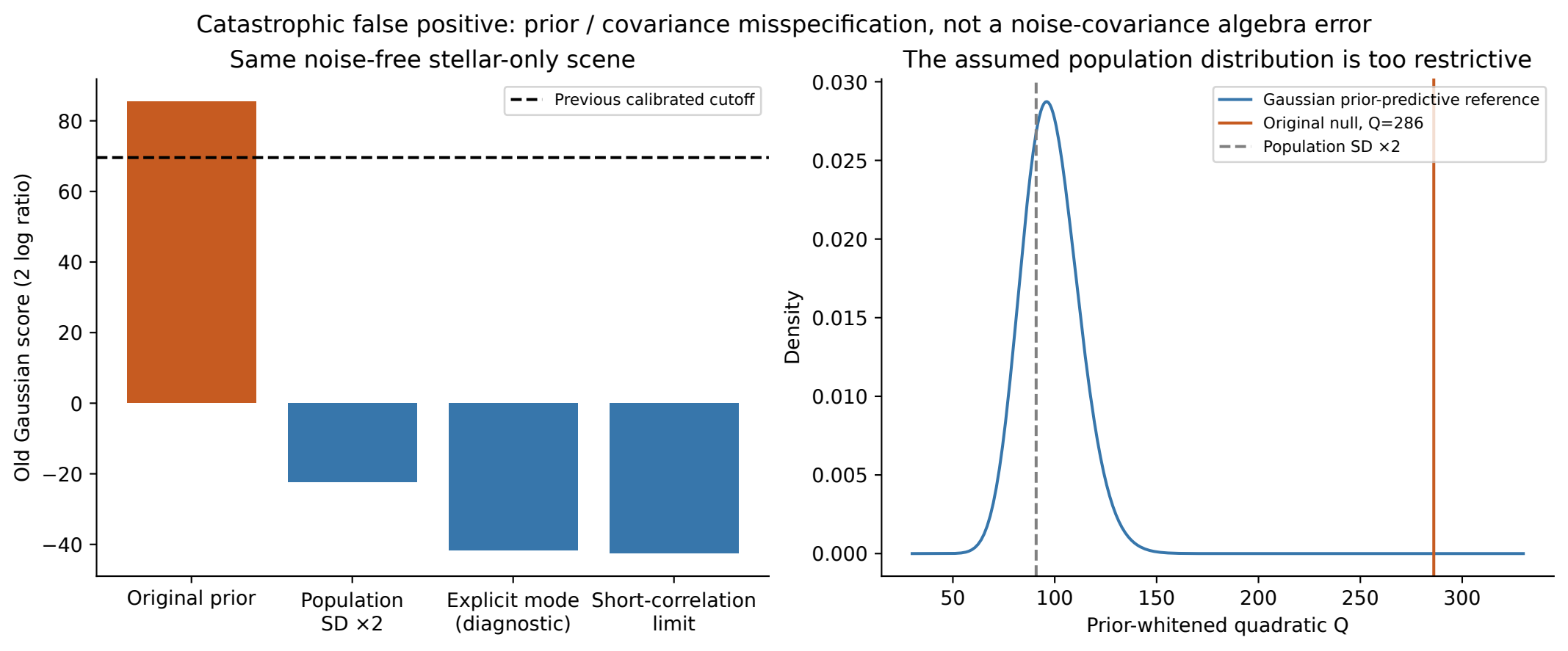
Catastrophic false positive: prior / covariance misspecification, not a noise-covariance algebra error
Same noise-free stellar-only scene
Previous calibrated cutoff
Old Gaussian score (2 log ratio)
80
60
40
20
0
−20
−40
Original prior
Population SD ×2
Explicit mode (diagnostic)
Short-correlation limit
The assumed population distribution is too restrictive
Gaussian prior-predictive reference
Original null, Q=286
Population SD ×2
Density
0.030
0.025
0.020
0.015
0.010
0.005
0.000
50
100
150
200
250
300
Prior-whitened quadratic Q